Question 1: (15 points total + 2 points extra credit)

a. (5 points) If a separate measurement shows that peak radiance emitted from the lava is at

1.6 microns, what is the radiant temperature of the lava (in Kelvin)? Assume lava is a

Blackbody.

Using Wein's Displacement Law, I found that the lava temperature will be about 1811.25K in this scenario.

b. (5 points) What temperature would an object need to be to appear brightest in color IR

film (assume the peak sensitivity of color IR film is at 0.80 microns)?

The object would need to be about 3622.5K

c. (5 points) Explain (in terms of the physical characteristics of the scene) some of the

temperature variability that you can see in the Mt. Etna thermal image. Use the

multispectral false-color image to understand the land-cover features on and around Mt.

Etna and how they affect what you see in the thermal image. Hint: Answer the question in

terms of relative temperature - no need to compute specific temperatures around the image.

In the thermal image, the brightest colors are expressing the highest radiance thus the highest temperature. The right (east?) side of the mountain is almost pure white telling me that there is a high heat signature in that area. The blotchy dark black that is pouring out of the left (west?) appears to be smoke, blocking any thermal signature that could be coming from the surface. Around the edges of the mountain are dark patches that appear to be forested landcover, expressing cooler temperatures than where the eruption or bare rock is.

Thinking ahead (2 points extra credit): You've produced a false color image. What do the different colors tell you? Answer in terms of the strength of the spectral reflectance signal from each band, not what the signal tells you about the surface (we don't know that, yet). Focus on one or two of the colors that you perceive.

The different colors tell me where I have the strongest signal in that channel. The deeper red colors tell of a strong signal in the red band and weaker signals in the other bands. The pattern is similar respective to each color and band selection.

Question 2: (20 points total & 1 point extra credit)

a. (3 points) What are the units for radiance (i.e. the pixel values)? What are the units for wavelength (i.e. the band values)?

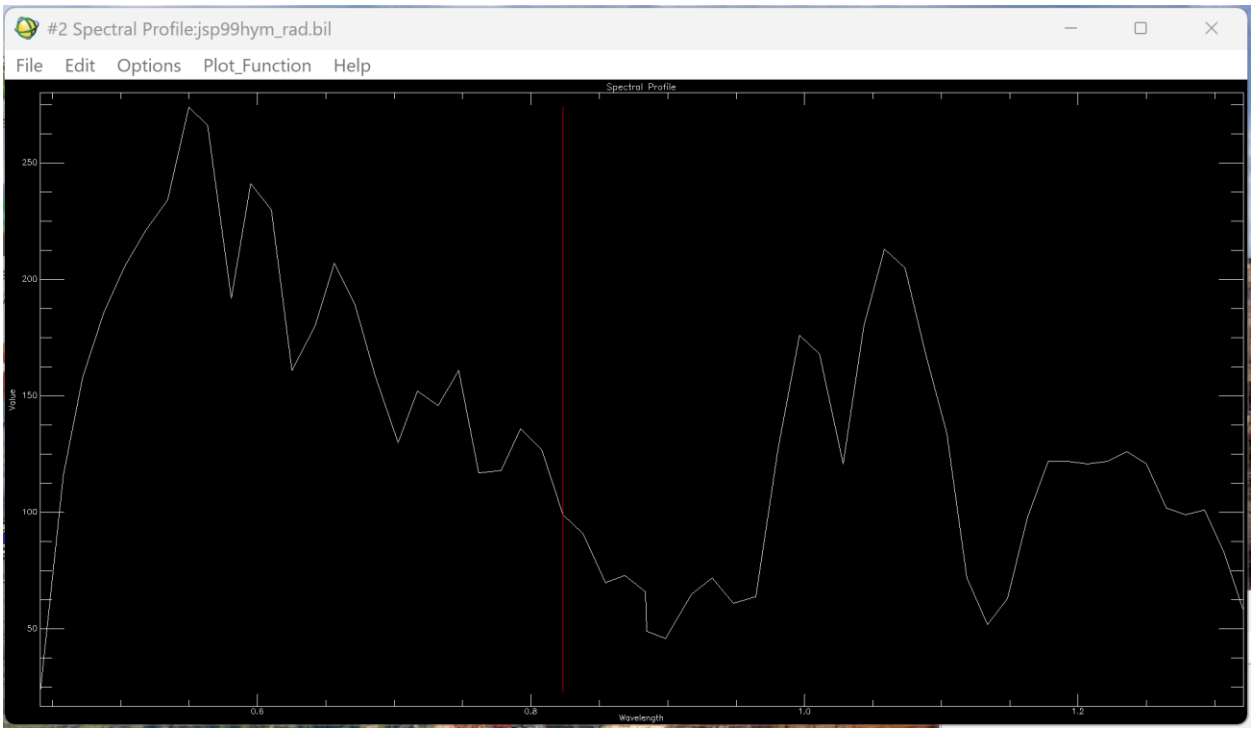
The units for radiance are watts per meter squared per steradian per micrometer. The units for wavelength are micron, or micrometer, or µm.

b. (3 points) Name the parts of the electromagnetic spectrum that are included in this spectral range.

The spectral range for this pixel ranges from ~0.4µm to ~1.4µm. The parts of that range include the visible and infrared.

c. (7 points) Which channel (wavelength) has the highest value for the sampled pixel? Explain the reasons why and include a screen shot of the profile/image to support your Answer.

The wavelength with the highest value at this pixel is 0.5496µm; in the green spectrum. The reasons why are because of the peak wavelengths that the sun emits, the absorption and reflectance of the lake, and the interactions with any sediments or algae in the lake.



d. (7 points) What are the causes of some of the atmospheric absorption features (the dips in the Z-Profile) that you can see in the spectrum of this pixel? Hint: see the textbook and lecture notes to recall which atmospheric components cause absorption at different wavelengths. Not all of the absorption features are due to the atmosphere so do not worry about the ones that don't match with well-known atmospheric absorption features.

Examining Figure 4.26 on p. 251 in the textbook, it appears that the absorption features in this spectrum are from atmospheric oxygen at about 750nm and water vapor in the 800-900nm and 1300nm range.

Thinking ahead (1 points extra credit): After we direct you to the pixel at Sample 246, Line 300 we state that, "This is one of the darkest pixels in the image and nearly all of the

measured radiance is from the atmosphere." How might you use this information to correct

for "atmospheric path interference".

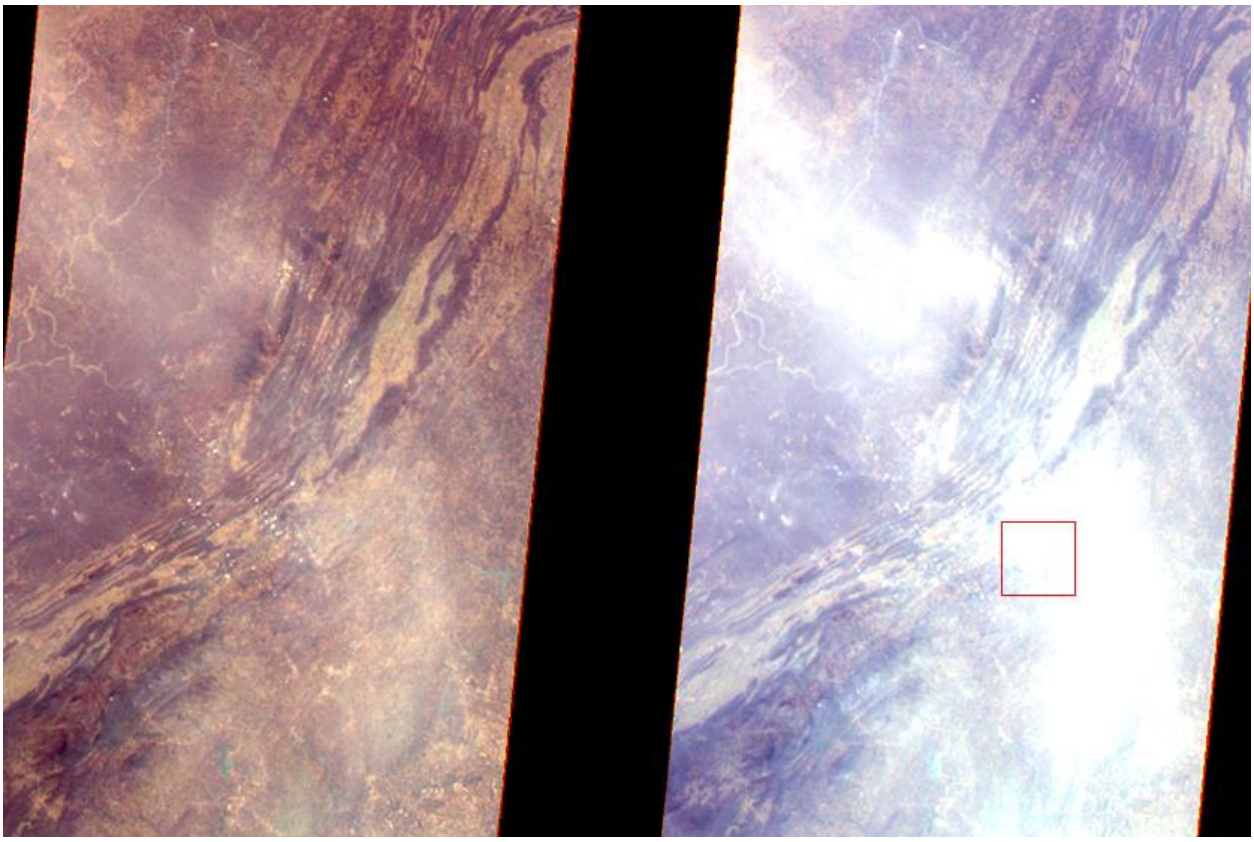
If we know that area is the darkest in the image, we can assume the sensor shouldn't be detecting there, or we can assume that what it is detecting is atmospheric. We can factor in a correction for pixels in that same band subtracting atmospheric quantities accros the imagery.

Question 3: (15 points total)

a. (8 points) What types of atmospheric scattering are you seeing in these images? Please

include a screen shot of images to help explain your answer.

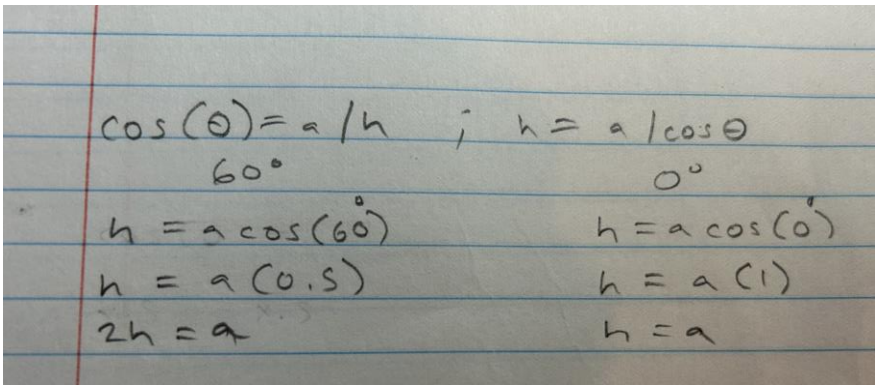
In this image, I see hazy white scattering that gets stronger at higher angles. This is likely due to mostly Mie and perhaps nonselective scattering. Since it is white, it tells me that all wavelengths are scattered, meaning that Rayleigh scattering can be (mostly) ruled out. The higher scattering in the higher angle images is due to more atmospheric interference since there is more atmosphere for the wavelength to pass through at these angles.



These images show the scattering at increased angles well.

b. (7 points) Compute how many times longer a 60-degree path length is compared to the 0 degree path length. Show your work for full credit.

Using the cosine effect, I calculated that the 60 degree path is 2 times longer than the 0 degree path.



IV. Problem Set (50 points total)

1. (4 pts) Develop a definition for "remote sensing" that is meaningful to you and relevant to your academic interests. Be concise and precise.

Remote sensing describes an array of techniques to help create and analyze images of the various surfaces and landcovers of Earth and atmosphere from afar using technology sensitive to various electromagnetic frequencies.

2. (16 pts) Fill in the following table. Watch your units (try to use units that are most appropriate for the magnitude of the number).

Giga = 1,000,000,000

mega = 1,000,000 (e.g. MHz = 1,000,000 Hz)

kilo = 1,000 (e.g. kHz = 1000 Hz)

giga = 1,000,000,000 (e.g. GHz = 1,000,000,000 Hz)

centi = 1/100 (e.g. cm = 1/100 m)

milli = 1/1000 (e.g. mm = 1/1000 m)

micro = 1/1,000,000 (e.g. µm = 1/1,000,000 m)

nano = 1/1,000,000,000

Wavelength	**Frequency**	**Name of the spectral region**
1.0 mm	~300Ghz	**microwave**
12.0 µm	~25 THz	infrared
1.6cm	19 GHz	microwave
50µm	6 x 10^{12} Hz	infrared
460 nm	650 THz	Visible
0.866 µm	346Thz	infrared
3.5mm	85 GHz	microwave
217	1380 kHz	radio

3. (16 pts) What kind of scattering (Rayleigh, Mie, or non-selective) would you expect to be most important when radiation of the specified wavelength encounters the following natural or anthropogenic particles? Hint: see the lecture slides and textbook for typical ran

Rayleigh : particle is smaller than wavelength.

Mie: particle is ~ same size as wavelength.

Nonselective: particle is larger than wavelength.

Wavelength	**O_2 molecules (size 10^{-10}m)**	**Smoke particles (size 0.3 µm)**	**Cloud droplets (size 20 µm)**	**Rain droplets (size 3 mm)**
550 nm	Rayleigh	Mie	nonselective	nonselective
11 µm	Rayleigh	Rayleigh	Mie	nonselective
1600 nm	Rayleigh	Rayleigh	nonselctive	nonselective
1 cm	Rayleigh	Rayleigh	Rayleigh	Mie

4. (14 pts) What is the radiant temperature of an object that has a true kinetic temperature of 300 K and an emissivity of 0.97? Show your work for full credit. What type of material might have this combination of temperature and emissivity? Hint: You'll find some help by reading the textbook on pgs. 253-4, in particular equation 4.16.

Radiant temperature of an object with kinetic temperature of 300K and emissivity of 0.97 is about 297.72K.

According to Table 4.4 in the book, the material that this aligns most closely with vegetation.

Equation 4.16

$$T_{rad} = \varepsilon^{1/4} \cdot T_{kin}$$

$$T_{rad} = (0.97)^{1/4} \cdot 300K$$

$$T_{rad} = 297.72K$$